# Akanksha Sachan

**Contact**

Joint Carnegie Mellon University-University of Pittsburgh PhD Program | aks203@pitt.edu
Aging Institute & Dept. of Immunology, School of Medicine, UPitt | LinkedIn
**Machine learning for Genomics & Epigenetics** | Twitter | Github | Google Scholar

## Education

**University of Pittsburgh** *'24 – Present*
*PhD Candidate in Computational Biology*
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

**Carnegie Mellon University** *'22 – '24*
*PhD Student in Computational Biology*
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

**Indian Institute of Technology (IIT), Bombay** *'18 – '22*
*BTech. in Chemical Engineering* | GPA: 8.06/10

## Awards and Honors

- **American Federation for Aging Research** (AFAR) Graduate Scholarship *'25*
- **Best Poster** Award at the Aging Institute Research Day, UPitt *'24*
- **1st Place, Student A100 track** NeurIPS LLM Efficiency Challenge | team of 5 *'23*
- **$500 Track Prize** (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 *'22*
- Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay *'22*
- Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship *'21*
- Awarded a full scholarship ($5,000) for iSURE undergraduate research program by the University of Notre Dame *'20*
- **Top 0.2%** (Rank 3188 out of **1.1M+** candidates) in JEE (Main) *'18*
- **Top 1.5%** (Rank 2517 out of **160k+** candidates) in JEE (Advanced) *'18*

## Invited Talks

**"FIREFate: Functional and Interpretable Regulatory Encodings of cell Fate"** | *paper-in-prep*

- Keystone Symposia: Artificial Intelligence in Molecular Biology *Sep '25*
- International Society for Computational Biology (ISCB)/(ISMB) *Jul '25, '26*
- American Association of Immunologists (AAI) Annual Meeting 2026 *Apr '26*
- Center of Lung Aging and Regeneration, School of Medicine, UPitt *Nov '25*
- 30th Annual School of Medicine Graduate Student Research Symposium, UPitt *Oct '25*
- Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat *Aug '25*

**"Metabolic rewiring of skeletal muscle driven by loss of DNA-repair"** | *paper-in-prep*

- Center for Systems Immunology Annual Retreat, UPitt *Nov '25*
- 23rd Annual Department of Immunology Scientific Retreat, UPitt *Sep '25*
- Aging Institute Research Day, UPitt *Nov '24*

**"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"**

- 50th European Calcified Tissue Society (ECTS) Congress *Apr '23*

## Athletic Accolades

**Bronze Medal** | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*
**3rd Position** | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*
**Swam 12-hours** | Swimathon organized by the IIT Bombay aquatics committee *'19*
**Ran 26.2 miles** | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*
**Ran 13.1 miles** | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

## Leadership & Societal Contributions

**Vice President** | *Allegheny Science Policy and Governance (ASPG)* *Aug '25 – Present*

- Leading a recognized chapter of the **National Science Policy Network**
- Bridged scientists and PA legislators on **AI/data-center and aging policy** by organizing discussions at the State Capitol, Harrisburg, PA

**Teaching Assistant** | *02-760: Lab Methods for Computational Biologists, SCS, CMU* *Fall '23 – Spring '24*

- Taught a lecture on **genomics technologies** and gene regulation to a class of around 20 students

**Coordinator** | *Technights, Carnegie Mellon University*, Robotics Institute *Aug '23 – Aug '25*

- Led a **computer-science outreach** program teaching middle-school girls technical sessions on data analysis, foundation models, and deep learning

**Team Lead** | *4D Nucleome Consortium Hackathon*, Seattle, Washington *Mar '24*

- Led a team of 5 during the hackathon to create integrative pipelines for **single-cell HiC** data analysis using deep-learning tools and publicly available cell-atlas scale datasets
- Delivered a 30-minute presentation to showcase results to consortium researchers

**Treasurer • Senator** | *Graduate Student Association, (CPCB GSA)* *Aug '23 – Aug '25*

- Managed a budget of **$5,121** and tracked spending spanning both CMU and UPitt GSA events
- Organized the 2024 open-house event for visiting prospective students admitted to CPCB

**Editorial Board Member • Design Convenor** | *Insight, the student newsletter of IIT Bombay* *'19 – '21*

- Sketched & Illustrated **1 Flagship** & **1 Special Edition Magazine Cover**, and a merch t-shirt
- Editorial Board member ('20-'21): contributed to editorial oversight and article curation

**Convenor, Girls Swim Team** | *Aquatics, Indian Institute of Technology Bombay* *'19 – '20*

- Led logistics, training coordination, and community building events for the institute girls' team

## Consortiums & Workshops

**SenNet Consortium Annual Meeting** | *Cellular Senescence Network, NIH, Washington, DC* *Sep '25*

- Attended talks on cellular senescence atlasing and organ-specific senescence signatures

**Grace Hopper Celebration** | *AnitaB.org, Philadelphia* *Oct '24*

- Experienced talks and brainstorming sessions with tech industry leaders

**Computational Genomics Summer Institute** | *University of California, Los Angeles* *Jul '23*

- Delivered Lightning Talk: "Connecting 3D genomic organization to gene expression"

**Grad Cohort for Women** | *Computing Research Association (CRA-WP), San Francisco* *April '23*

- Participated in mentoring sessions from women in CS across industry and academia

## Research Experience Prior to Thesis

**Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture** *'22 – '24*
*Supervisor – Prof. Jian Ma, Computational Biology Department, School of Computer Science, CMU*

- Developed a framework to identify cis-regulatory, 3-D genomic hubs, controlling target-gene expression and key transcription factors in healthy and cancerous cell lines
- Implemented **hypergraph-based representation learning** models for integrative analysis of 3D genome architecture in single-cell, multi-omic, mouse brain datasets

**Translational Mechanobiology Lab**, MBI, National University of Singapore *'22*
*Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine*

- Implemented a **Generative-Adversarial-Network** (GAN) pipeline for colorectal cancer imaging data to support robust deep-learning deployment in clinical environments
- Reviewed domain adaptation and distribution-shift methods for liver fibrosis staging from annotated medical images

**Lab for Bone Biomechanics**, Institute of Biomechanics, ETH Zurich *'21*
*Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech | Presentation*

- Added a **Type-2 Diabetes** pathway to an agent-based **osteoporosis** model and ran large-scale simulations on the **CSCS Swiss National Supercomputing Center Cluster**
- Validated the model against in-vivo data from 235 diabetic patients and identified osteoid production as a key parameter via **sensitivity analysis**

**Multicellular Systems Engineering Lab**, University of Notre Dame *'20*
*Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering | Poster*

- Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs using OpenCV and Fiji to understand organism-wide signaling patterns
- Evaluated variance under gene-knockout in the $Ca^{2+}$ signaling pathway using PCA-features

## Key Coursework

**Machine Learning & Statistics:** Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

**Predicting genomic-hub mediated TF-perturbation response via geometric deep learning** *'23*
*10-708: Probabilistic Graphical Models | Code Repo*

- Augmented the GEARS GNN for Perturb-seq response prediction with a Hi-C contact graph (K562 **cancer cell-line** Micro-C bulk-seq dataset, 20kb (base-pair) bins mapped to gene TSSs), adding 3D-genome topology to the existing co-expression and Gene Ontology graph features

**Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs** *'23*
*02-710: Computational Genomics | Code Repo*

- Developed a **GNN** to predict methylation from HiC data, connecting structure-to-phenotype
- Performed clustering with K-means and t-SNE on **single-nucleus DNA methylation** data

**Efficient Online Continual Semi-Supervised Learning** *'22*
*10-701: Intro to ML*

- Improved semi-supervised OCL-PDS with active learning and optimized pseudo-labeling
- In the Amazon-WPDS dataset, our approach outperformed the ER-FIFO method